

Integrated Analysis of a Petroleum Field Under Drive Mechanisms Conditions

1. Introduction

The performance of a petroleum reservoir relies not only on fluid properties and its geological characteristics but also on the energy that controls hydrocarbon production. This energy is provided by mechanisms that influence pressure behaviour and recovery potential.

Understanding the performance, production, and nature of hydrocarbon fields is essential in petroleum engineering in order to determine the recovery efficiency and the overall behaviour of the reservoir throughout its productive life. A proper evaluation allows engineers to forecast pressure decline, estimate production potential, and design strategies to maximise hydrocarbon recovery.

This project evaluates a conceptual petroleum reservoir to analyse the influence of different drive mechanisms on reservoir behaviour, recovery efficiency, and long-term production outcomes.

2. Nomenclature

A	Reservoir area [m ²]
h	Reservoir thickness [m]
GRV	Gross Rock Volume [m ³]
NTG	Net-to-Gross ratio [–]
ϕ	Rock porosity [–]
Sw	Initial water saturation [–]
Bo	Volumetric factor of petroleum [–]
STOOIP	Stock Tank Original Oil In Place [m ³]
P0	Initial reservoir pressure [kPa]
Pwf	Bottom-hole flowing pressure [kPa]
Pf	Future pressure in reservoir [kPa]
Q	Initial flow rate [m ³ /s]
J	Productivity Index [m ³ /s/kPa]

3. Methodology

A Python-based model was implemented to automate reservoir calculations, recovery estimation, and comparative scenario analysis by considering its geological characteristics, fluid properties, and drive mechanism to determine its overall performance.

3.1 Theoretical Background

The procedure begins with the calculation of the Gross Rock Volume (GRV), which represents the total bulk volume of geological structure above hydrocarbons.

$$GRV = A * h \quad (1)$$

Additionally, the Stock Tank Original Oil In Place (STOOIP), which represents the total volume of crude oil located in the reservoir before its extraction.

$$STOOIP = \frac{GRV * NTG * \phi * (1 - S_w)}{Bo} \quad (2)$$

Furthermore, the productivity index measures the efficiency of the well to produce fluids.

$$J = \frac{Q}{P_0 - P_{wf}} \quad (3)$$

The last subsection outlines the estimation of flow rate in the reservoir over time.

$$Q = J * (P_f - P_{wf}) \quad (4)$$

3.2 Parameters

TABLE 1. PARAMETERS AND VALUES USED IN SIMULATION

Parameter	Value	Units
Area	6.07×10^6	m ²
Thickness	21.3	m
Net-To-Gross	0.8	–
Water saturation	0.21	–
Volume factor	0.25	–
Porosity	0.12	–
Initial water pressure	26200	kPa
Final water pressure	13800	kPa
Initial flow rate	95.4	m ³ /s

The parameters and values applied during this simulation are listed in Table 1. For the purpose of this study, a future reservoir pressure of 20000 kPa is assumed.

TABLE 2. RECOVERY FACTORS

Drive Mechanism	Recovery Factor Range	Recovery Factor Applied
Rock & Fluid Expansion	1 - 5%	5 %
Solution Gas Drive	5 - 30%	20 %
Gas Cap Drive	20 - 40%	30 %
Gravity Drainage	50 - 80%	40 %
Aquifer Drive (Weak)	10 - 35%	35 %
Aquifer Drive (Strong)	35 - 60%	45 %

In terms of drive mechanisms, Table 2 shows both recovery factor ranges and applied during the simulation. They represent the extraction efficiency per drive mechanism.

4. Results

TABLE 3. RESULTS AFTER SIMULATIONS

Drive Mechanism	GRV [m ³]	STOOIP [m ³]	Productivity Index [m ³ /s/kPa]	Recoverable Oil [m ³]	Estimated Flow Rate [m ³ /s]
Rock & Fluid Expansion	129291000	13575555	0.0077	678777.75	47.7
Solution Gas Drive	129291000	13575555	0.0077	2715111	47.7
Gas Cap Drive	129291000	13575555	0.0077	4072666.5	47.7
Gravity Drainage	129291000	13575555	0.0077	5430222	47.7
Aquifer Drive (Weak)	129291000	13575555	0.0077	4751444.25	47.7
Aquifer Drive (Strong)	129291000	13575555	0.0077	6108999.75	47.7

As shown in Table 3, all parameters remain constant except for the volume of oil recovered. This consistency reflects the lack of variation in geological characteristics such as rock volume and porosity within the reservoir. In contrast, the recoverable oil volume changes according to the recovery factor assigned to each drive mechanism.

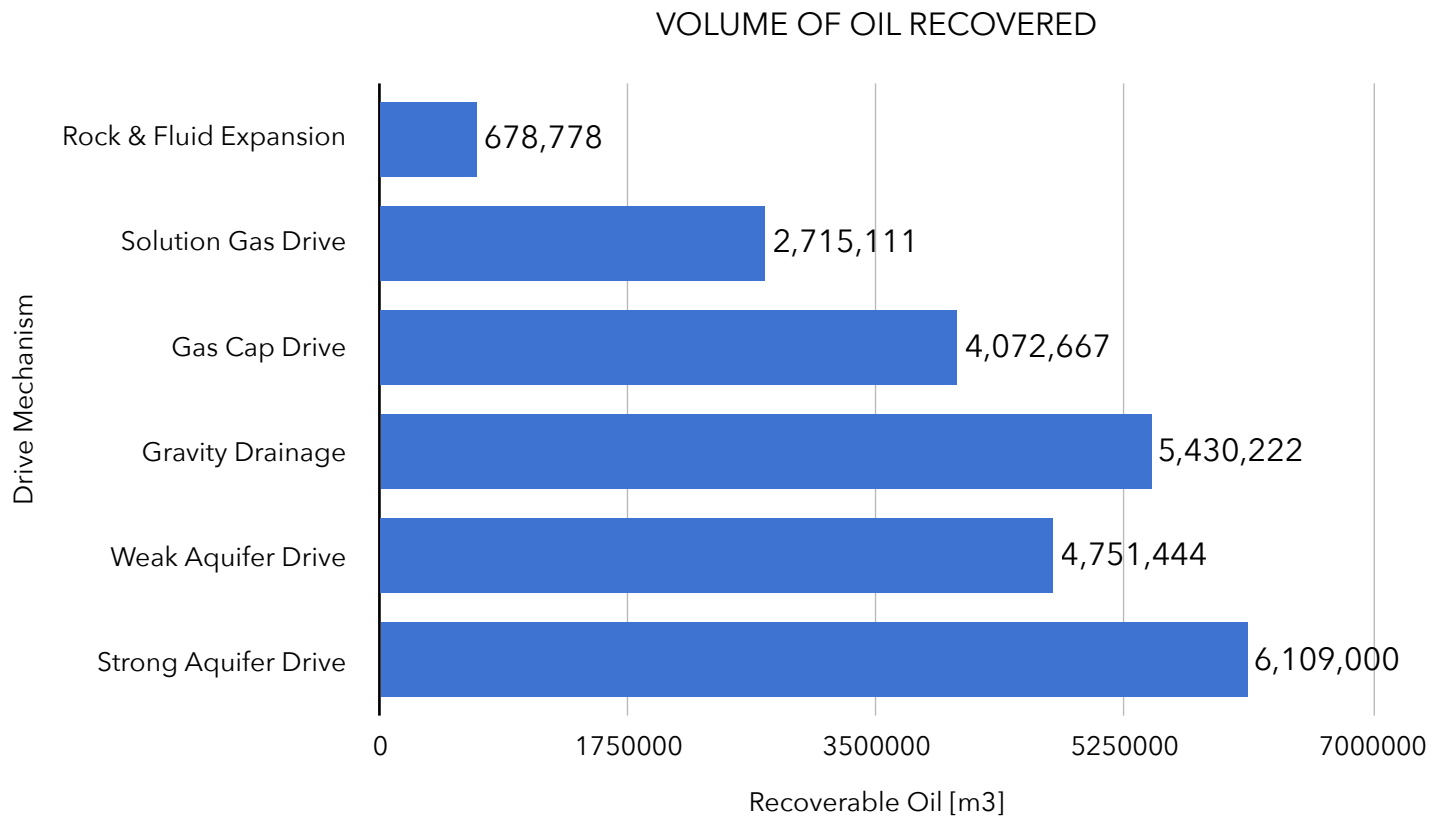


FIGURE 1. VOLUME OF OIL RECOVERED

Figure 1 presents the comparison of drive mechanisms based on the volume of oil recovered from the same reservoir, illustrating the direct relationship between recovery factor and recovered volume. Rock and fluid expansion results in 678,777.5 m³, the lowest value obtained in the simulations. In contrast, the strong aquifer drive yields the highest recovered volume, reaching 6,108,999.75 m³.

5. Model Limitations

This model assumes constant reservoir properties and applies representative recovery factors for each drive mechanism. Therefore, the results should be interpreted as a conceptual comparison between each drive mechanism rather than a full dynamic reservoir simulation.

6. Conclusion

The evaluation of the petroleum field demonstrates how strongly reservoir performance depends on the dominant drive mechanism. Although all geological and fluid parameters remain constant across the simulations, the volume of recoverable oil varies significantly depending on the differences between recovery factors. Rock and fluid expansion provides the weakest energy source, while strong aquifer support yields the highest recovery due to sustained pressure maintenance and improved displacement efficiency.

By comparing these mechanisms under identical reservoir conditions, the study highlights the critical role of reservoir energy in determining production potential and long-term performance. This interactive analysis reinforces the importance of understanding drive mechanisms when forecasting recovery, selecting development strategies, and making informed engineering decisions in real-field scenarios.

7. Reference

Petroleum Engineering, Principles, and Concepts CPD Materials.